

Alexander Koroshchenko

Senior Gameplay Programmer

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Professional Summary

Performance-driven Senior Gameplay Programmer with over 14 years of experience in game development, specializing in gameplay systems, VR mechanics, and real-time optimization. I have a strong background in physics and math, C++ and Unreal Engine. Passionate about game development and programming in general, curious about technologies. Have shipped projects for PC, consoles, mobile and VR. And my favourite CPU is 6502.

Skills

Languages: C++, C, Objective-C, Swift, Zig, Go, Python, x86 assembly

Engines/Tech: UE4/UE5, ImGUI, Cocos2d, Box2D, Oculus Rift, VR, SwiftUI, UIKit, Git, Perforce, SVN

Professional Experience

Senior Gameplay Programmer | Build A Rocket Boy — *Edinburgh, UK*

Jul 2020 – Present

- Developed and maintained single-player and multiplayer gameplay systems, including, but not limited to:
 - Open-world destruction (explosive gameplay elements, various destructible props)
 - Vehicles and robots destruction and VFX systems
 - Inventory system (in collaboration with backend and UI teams)
 - Story dialogue system used in cinematics and game missions
- Created internal development tools to support design iteration and debugging
- Performed profiling and various optimizations
- Conducted code reviews and participated in technical interviews.

Links:

<https://www.youtube.com/watch?v=fHP4LRdUIYk>

<https://www.youtube.com/watch?v=90FNVadJP6s>

Senior Gameplay Programmer | HYVRLabs — *Remote*

February 2017 - September 2019

- Converted most of the project from BP to C++

- Implemented key gameplay systems such as:
 - Hit-scan weapons
 - VR player character and hand interaction systems
 - AI logic (behavior system, cover mechanics, enemy classes, etc.)
 - UI systems (health bars, weapon indicators)
- Executed performance and memory optimizations for VR
- Managed a small team of programmers and artists
- Conducted code reviews and participated in technical interviews

Links: https://store.steampowered.com/app/1135850/NearPrime_Net_Burn/

Game Developer | Freelance — Remote

December 2013 – November 2017

- Worked with Lorenzo Carver (<https://www.linkedin.com/in/lorenzocarver>) on prototype applications exploring gamification concepts:
 - Mobile arcade racing game with social networking elements and procedural 2D terrain
 - Business simulation game for calculating metrics like LTM, NTM, and cash flow
 - Voice analysis app for iOS and WatchOS (real-time pitch, timbre, and WPM detection)
- Participated in multiple hackathons, both remotely and on-site
- Co-founded an indie game startup with four colleagues:
 - Developed a sci-fi RPG with procedurally generated, destructible environments
 - Designed and implemented inventory, UI, and AI systems (e.g., different modes, weapons, difficulty levels)

Game/Tools developer | Lemon Tree Studio and other projects — Omsk, Russia

August 2008 – December 2013

- Developed various mobile games and apps for iOS and macOS using Cocos2d and Box2D
- Mentored intern programmers, helped them with game development frameworks
- Participated in technical interviews and toolchain development

Education

MSc in Solid State Physics — 2005-2011

Omsk State University, Russia

Performance-Aware Programming Course — Present

Casey Muratori's course on performance at computerenhance.com